Resilience Academy

SDI System Installation & Customization with CRD and KoboToolbox

Theme 1: General Introduction

Development Environment & Operations Fundamentals

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Day 1 of 4 | crd.resilienceacademy.ac.tz

Resilience Academy

Day 1 Overview

Goal: Lay the DevOps foundation for all SDI sessions

Every participant leaves Day 1 with a working environment and having accessed live CRD data.

Morning — Lectures

- SDI concepts & CRD introduction
- Required software stack
- Server environments (local, VM, cloud)
- DevOps: Git, Docker, Docker Compose

Afternoon — Practicals

- 1.1: Install dev environment (3h)
- 1.2: Linux server navigation (3h)
- 1.3: Git, Docker, Compose hands-on (3h)

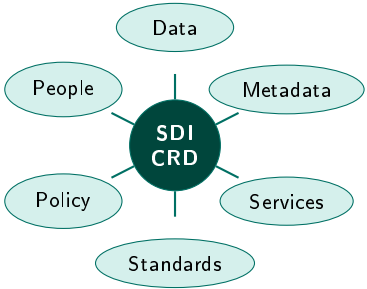
Live Reference Platform

`crd.resilienceacademy.ac.tz` | Mwanza: `mwanza_businesses`, `mwanza_education_facilities` | Tanga: `Tanga_imagery_reduced`

What is a Spatial Data Infrastructure?

A **framework** of policies, standards, technologies and people enabling geospatial data sharing.

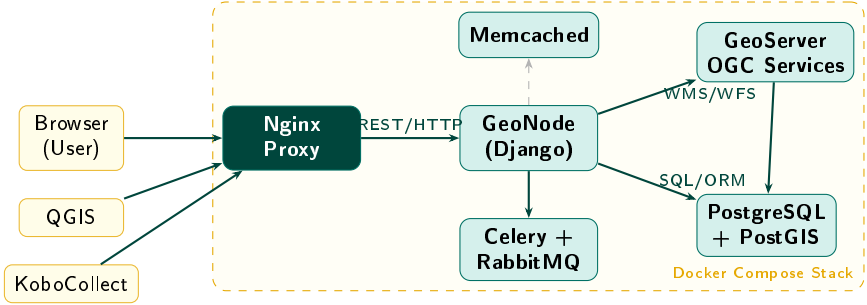
- **Data** — vector and raster datasets
- **Metadata** — ISO 19115 descriptions
- **Services** — WMS, WFS, WCS, CSW
- **Standards** — OGC, INSPIRE
- **People & Policy** — governance



CRD = Tanzania's SDI

GeoNode platform. 287+ climate risk datasets.
Partnership: UTU, Ardhi, SUZA, SUA, GIZ.

CRD Architecture Diagram



OGC Services: automatically available for every layer uploaded to CRD

WMS (maps) **WFS** (vector) **WCS** (raster) **WMTS** (tiles) **CSW** (metadata)

Software Stack & Deployment Pipeline

Required Software

Tool	Role
Docker + Compose	Containerisation
Git	Version control
VS Code	Code editor
Python 3 + GeoPandas	Analysis
QGIS	GIS validation
Postman	API testing
pgAdmin	Database admin

Hardware

8GB RAM · 20GB disk · Ubuntu 22.04 or WSL2

Dev to Production Pipeline

Local Dev → VM Stage → Cloud VPS → CRD Live

Key Ports

22 SSH · 80 HTTP · 443 HTTPS
5432 PostgreSQL · 8000 GeoNode
8080 GeoServer · 6379 Redis

DevOps: Git + Docker + Compose

Git Commands

```
git clone <repo> # get repo
git status # see changes
git add . # stage all
git commit -m "msg" # save
git push origin main # share
git pull # sync
git log --oneline # history
```

Docker Commands

```
docker --version
docker compose version
docker run hello-world
docker ps # running
docker logs -f <name>
docker stop/rm <name>
docker images
docker stats
```

CRD Quick Start (Day 2)

```
git clone https://github.com/GeoNode/
  geonode-project.git my-crd
cd my-crd && cp .env.sample .env
docker compose build
docker compose up -d
docker compose exec django \
  python manage.py createsuperuser
```

Compose Commands

```
docker compose up -d # start
docker compose down # stop
docker compose ps # status
docker compose logs -f # logs
docker compose exec django bash
```

Essential Linux Commands

Files & Navigation

```
pwd ; ls -la # where / list
cd /path ; cd ~ # navigate
mkdir dir # create dir
touch f.txt # create file
cp src dst # copy
mv src dst # move/rename
rm f.txt # delete
cat f.txt # view
nano f.txt # edit
sudo cmd # superuser
```

System & Networking

```
top ; free -h ; df -h # monitor
ip addr show # interfaces
ss -tlnp # ports
ping crd.resilience...ac.tz

# Query live CRD API
curl "https://crd.resilienceacademy.ac.tz
  /api/v2/datasets/?search=mwanza" \
  | python3 -m json.tool | head -20
```

Day 1 Summary

Achieved

- ✓ SDI concepts + CRD platform understood
- ✓ Software stack installed and verified
- ✓ Server environments understood
- ✓ Git, Docker, Compose mastered
- ✓ CRD API queried from terminal
- ✓ Multi-container app running locally

Day 2: CRD Installation

- GeoNode architecture deep-dive
- CRD Docker Compose installation
- CRD Ubuntu bare-metal installation
- Customise branding and settings
- Load Tanga & Mwanza datasets

Ensure: `docker compose version` works!

Questions?

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